

Prompt Hint Library - Reusable Amazon Q Hints for Custom Modernizations

1. Controller Modernization

```
// Q Hint: Convert this ASP.NET MVC controller to ASP.NET Core Web API.  
// Use attribute routing, constructor injection, and IActionResult return types.  
// Assume the services are registered in DI and follow async pattern.
```

2. Dependency Injection

```
// Q Hint: Replace direct instantiation with constructor-based dependency  
injection.  
// Assume ILogger<T> and IStudentService are available via DI.
```

3. Data Access – ADO.NET to EF Core

```
// Q Hint: Replace SqlConnection and SqlCommand usage with EF Core DbContext.  
// Assume DbContext has a Students DbSet with properties Id, Name, Age.
```

4. Synchronous to Asynchronous

```
// Q Hint: Convert all synchronous DB calls to async/await using EF Core.  
// Use Task-based method signatures.
```

5. HttpContext / Session Modernization

```
// Q Hint: Replace HttpContext.Current usage with IHttpContextAccessor.  
// Refactor to support scoped services and ASP.NET Core request lifecycle.
```

6. Configuration

```
// Q Hint: Replace ConfigurationManager.AppSettings with IConfiguration and  
strongly-typed settings classes.  
// Assume config values are loaded from appsettings.json.
```

7. Logging

```
// Q Hint: Replace System.Diagnostics.Trace or Console.WriteLine with ILogger<T>  
from Microsoft.Extensions.Logging.
```

8. Routing & API Attributes

```
// Q Hint: Add [ApiController] and [Route("api/[controller]")] to modernize  
routing conventions.
```

9. Startup & Middleware

```
// Q Hint: Remove Global.asax and replace with ASP.NET Core middleware-based  
startup pattern.  
// Assume a Program.cs with WebApplication.CreateBuilder().
```

10. Validation

```
// Q Hint: Replace manual model validation with Data Annotations and  
[ApiController]-enabled model binding.
```